# AI-Based Groundwater Depletion Risk Prediction for Mysuru District Using Machine Learning

*Mohammed Sadiq Shariff* *Shumail Ali* *Deeksha Prasad S P* *Nagashree D*
Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)
Maharaja Institute of Technology, Thandavapura
Mysuru, Karnataka, India
Corresponding Author: mohammedsadiqshariff80@gmail.com

***Abstract*—Groundwater depletion is a critical environmental threat in semi-arid regions of peninsular India. This paper presents an exploratory machine-learning framework for predicting groundwater level (GWL) and classifying depletion risk at manually monitored wells in Mysuru district, Karnataka. We assemble longitudinal groundwater records from official CGWB archive snapshots via the National Water Data Portal (NWDP)—historical archive (1991–2020) and recent archive (2021–2025)—yielding 4,185 modelling records (1997–2023) across 92 stations in Mysuru district paired with daily district-averaged rainfall from the NWDP Karnataka daily rainfall series. A Historical Gradient Boosting Regressor (HistGBT) trained on features derived from GWL lags, rolling means, station-historical statistics, and rainfall accumulation windows predicts the *next* observed GWL (horizon: 40–200 days, median 91 days) with MAE = 1.94 m, RMSE = 2.96 m, and $R^2 = 0.547$ on 657 held-out test records (November 2019–November 2023). A companion Random Forest classifier assigns each observation to one of four depletion-severity classes—No Depletion, Low, Moderate, and High—achieving 52.8% accuracy and macro-F1 of 0.457, outperforming majority-class (45.8%) and stratified-dummy (30.4%) baselines. High-risk episodes remain the hardest to identify (F1 = 0.33). An interactive web dashboard is provided for exploratory visualisation; it is not intended as an operational early-warning tool or a substitute for field measurements. All results are traceable to archived prediction artefacts and require independent hydrogeological validation before operational deployment.**

***Index Terms*—groundwater level prediction, depletion risk classification, machine learning, ensemble methods, Mysuru district, CGWB, temporal validation, exploratory dashboard**

## I. Introduction

Groundwater constitutes a vital natural resource for rural drinking water supplies and agricultural irrigation across peninsular India. In Karnataka's Mysuru district—a mixed agricultural and peri-urban region—seasonal rainfall variability, unregulated extraction, and heterogeneous aquifer lithology combine to produce sharp inter-annual fluctuations in the water table. Official CGWB assessments record *depth to water table* (metres below ground level, mbgl), where larger values indicate greater depth and therefore worse groundwater conditions.

**Research motivation.** Long monitoring gaps (typically 30–200 days, median 91 days) and the absence of continuous sensor networks make it difficult to detect emerging depletion trends in real time. A data-driven model that estimates the GWL at the time of the *next scheduled observation* could assist field staff in prioritising wells and allocating monitoring resources.

**Research questions.**

1) Can GWL at the next observation—a horizon of 40–200 days—be estimated from historical in-situ observations, station characteristics, and accumulated rainfall?
2) Can a magnitude-of-change label (No Depletion, Low, Moderate, High) be predicted reliably enough to support exploratory risk screening?

**Scope and limitations.** This study covers 92 CGWB monitoring stations within Mysuru district. It does not claim district-wide representativeness, live forecasting capability, or readiness for operational deployment. Predictions reflect patterns in historical quarterly monitoring data; they cannot replace hydrogeological field surveys or CGWB ground-truth assessments. The risk labels are defined empirically (Section IV-A) and require hydrogeological validation.

The remainder of the paper is organised as follows: Section II surveys related work; Section III describes datasets and preprocessing; Section IV presents the methodology; Section V details experimental setup; Section VI reports verified results; Section VII discusses limitations; Section VIII concludes.

## II. Related Work

Machine-learning methods for groundwater level (GWL) prediction have expanded rapidly, encompassing classical ensembles, hybrid deep architectures, and physics-informed models (Table I). Boomika *et al.* [1] developed a hybrid Random Forest–LSTM architecture combined with IoT-based monitoring for urban Chennai. Zeydalinejad *et al.* [2] evaluated ANN, LSTM, SVR, Random Forest, and wavelet-enhanced models across hydro-climatic inputs, highlighting performance variations across wells and seasons. Shuai *et al.* [3] integrated physical MODFLOW numerical simulation with deep LSTM networks to constrain predictions with hydrological dynamics. Sankaran *et al.* [4] employed signal decomposition and adaptive ensemble learning to forecast groundwater pressures in industrial zones of Tiruppur under climate variability. Guo *et al.* [5] integrated socioeconomic groundwater withdrawals into deep CNN and ResNet models to predict basin-scale groundwater storage in the Yellow River Basin.

**Research gap.** While these studies demonstrate predictive capability in dense sensor networks, physical simulations, or industrial basins, key limitations persist for regional semi-arid settings: (i) lack of localized depletion-risk classification paired with a decision-support dashboard; (ii) reliance on continuous telemetry or complex physical aquifer parameters that are unavailable in sparse manual monitoring networks; and (iii) absence of strict chronological validation isolating future records. Our work addresses these gaps using 27+ years of longitudinal CGWB data in Mysuru district, strict chronological splits (70/15/15), empirical depletion-severity categorization, and an interactive exploratory dashboard.

## III. Dataset and Preprocessing

### A. Groundwater Data

Groundwater level records originate from the Central Ground Water Board (CGWB) manual quarterly monitoring network published via the National Water Data Portal (NWDP) [6]. The underlying data correspond to two primary portal resources:

1) **Historical archive (1991–2020):** NWDP resource `3dacbc24`, archived as `gwl_manual_quarterly_cgwb_ka_1991_2020.csv` — quarterly manual well measurements for Karnataka with raw column "Groundwater Level Quarterly Manual (meter)" and timestamp "Data Acquisition Time" (`DD-MM-YYYY HH:MM`).
2) **Recent archive (2021–2025):** NWDP resource `7b76b032`, archived as `groundwater_karnataka_2021_2025.csv` — maintaining the same schema for recent monitoring years.

*Data provenance note*: While the official NWDP portal URLs provide the authoritative access point for live CGWB monitoring records, the experimental results herein are derived from the verified local training snapshots to guarantee full numerical reproducibility. Portal and local CSVs are not assumed to be strictly identical without a full live-pipeline revalidation, so historical record counts are preserved.

Records are filtered to Mysuru district (case-insensitive match on "mysore" or "mysuru"). After merging, deduplicating (on station, date, and coordinates), and removing impossible values (GWL $\leq 0$ mbgl or invalid dates), the dataset comprises **4,185 station-observation pairs** from **92 unique stations**. GWL is expressed in metres below ground level (mbgl); larger values indicate a deeper water table.

### B. Rainfall Data

Daily rainfall data correspond to the Karnataka manual daily rainfall dataset hosted on the National Water Data Portal (`nwdp.nwic.gov.in/.../rainfall-manual-daily-karnataka`) [7], assembled from a verified local snapshot of 15 multi-year daily CSV files (`rainfall_YYYY_YYYY.csv`, 1989–2025) and filtered to Mysuru district by name matching. Official portal datasets and local training snapshots are distinguished to ensure reproducible results. Daily values are averaged across all district gauges. A complete daily calendar is constructed between the earliest and latest available dates. Missing days within the calendar are *retained as* `NaN` and are not imputed to zero, because missing rainfall records and genuine zero-rainfall days are indistinguishable. Rolling windows (7, 30, 90, 180, and 365 days) are computed with `min_periods=1` so that early-series values are not wholly discarded, but the incomplete coverage of missing intervals is acknowledged as a limitation. Rainfall features are joined to groundwater observations using a backward `merge_asof` (nearest prior date), so no future rainfall information enters any feature.

*1) December 2023 Rainfall Gap:* Inspection of the assembled rainfall series identifies a data gap in the December 2023 period corresponding to the final months of the `rainfall_2023_2025.csv` file. This gap means that rolling accumulation features for groundwater observations in that window may understate actual rainfall. Because the test set extends to November 2023 and the affected observations are at the boundary, their rolling features carry partial-window uncertainty. This limitation is noted for future data-collection efforts.

### C. Feature Engineering

All lag and rolling features are computed *within station groups* using shift operations applied before the rolling window, ensuring that no observation contributes to its own feature values (no target leakage). The feature set (Table II) includes:

- **GWL lags**: $\ell_1$–$\ell_4$ (prior one through four observations at the same station).
- **GWL rolling statistics**: 3- and 4-observation rolling means of the preceding sequence; 3-observation rolling standard deviation.
- **GWL change features**: $\Delta_1 = \ell_0 - \ell_1$ and $\Delta_2 = \ell_0 - \ell_2$.
- **Station historical statistics**: expanding-window mean and standard deviation of all prior observations at the same station.
- **Rainfall accumulations**: 7, 30, 90, 180, and 365-day totals prior to each observation.
- **Temporal features**: year, month, quarter; cyclical month encoding (sin, cos).
- **Station identifiers**: Station name, Tehsil, Block, Village (one-hot encoded); latitude, longitude.

Records with fewer than four prior observations within a station (needed to populate all lag features) are removed (training input). At inference time, missing lag values are imputed by median (within the preprocessing pipeline), and the model handles structural missingness natively (HistGBT uses native NaN support).

## IV. Methodology

### A. Target Construction

**Regression target.** For each station-observation $t$, the target is the GWL recorded at the immediately subsequent observation $t+1$ at the same station:

$$y_t^{\text{reg}} = \text{GWL}_{t+1}$$

TABLE I
SUMMARY OF RELATED MACHINE-LEARNING STUDIES FOR GROUNDWATER PREDICTION

| Year | Authors | Paper | Method Used | Key Finding | Limitation / Research Gap |
|---|---|---|---|---|---|
| 2026 | S. Boomika *et al.* [1] | An AI-Driven Hybrid Random Forest–LSTM Framework for IoT-Based Groundwater Level Prediction and Sustainable Water Management in Chennai, India | Hybrid RF–LSTM with IoT sensing (depth, rainfall, temperature, soil moisture) | Outperforms standalone models on Chennai urban aquifer (RMSE = 0.38 m, MAE = 0.29 m, $R^2$ = 0.96), cutting LSTM RMSE by ∼22%. | Relies on real-time IoT sensor network and soil-moisture telemetry; lacks regional-scale categorical depletion risk screening and decision web tools. |
| 2026 | N. Zeydalinejad *et al.* [2] | Performance and Limitations of Machine Learning Models for Groundwater Level Prediction Using Hydro-Climatic Variables | ANN, LSTM, SVR, Random Forest, and wavelet transforms; introduced DDTI metric | Across 11 UK wells, validation $R^2$ varied widely ($< 0.1$ to $> 0.9$) by hydrogeology; 2-month lag optimal; sharp performance drop in summer ($R^2 < 0.1$). | Focuses on hydro-climatic sensitivity and model failure modes in UK basins; does not establish localized categorical depletion-risk screening. |
| 2026 | G. Shuai *et al.* [3] | Accurate Groundwater Level Predictions: Integration of Physics-Driven and Machine Learning Approaches | MOD-LSTM (coupling MODFLOW physical simulation with LSTM) | On 10 wells in North China Plain, 80% achieved NSE $> 0.75$; reduced RMSE by 0.618 m vs. MODFLOW and 0.280 m vs. pure LSTM; physical model aids interpretability. | Requires detailed hydrogeological parameters, boundary conditions, and MODFLOW grid inputs unavailable in data-sparse fractured rock. |
| 2026 | H. Sankaran *et al.* [4] | Assessing and Forecasting Groundwater Resources in the Context of Climate Change Using AI Techniques for the Industry Zones in Tiruppur, India | Signal decomposition (ICEEMDAN, VMD) + SMA-LSTM and CNN-LSTM with Adaptive Weighting | In Tiruppur industrial zone, ensemble achieved MAE = 0.267 m, RMSE = 0.429 m, $R^2$ = 0.948, NSE = 0.938; CMIP6 scenarios show future climate stress. | Complex signal decomposition and metaheuristic optimization tailored to industrial point wells; does not offer lightweight tabular screening or web UI. |
| 2026 | S. Guo *et al.* [5] | Integrating Socioeconomic Withdrawals Into a Deep Learning Framework for High-Resolution Groundwater Storage Prediction in the Yellow River Basin | Deep learning (CNN, ResNet) with groundwater balance and socioeconomic extraction constraints | Downscaled GRACE to 1 km across Yellow River Basin; balance constraints cut error by 28% (CNN) and 68% (ResNet); 81.2% of 101 wells had $r > 0.7$. | Predicts basin-scale satellite storage anomalies (GWSA) rather than local well-head depth-to-water table; requires socioeconomic withdrawal data. |

Note: Publication metadata, methods, and findings for cited 2026 studies were directly verified against the published journal articles.

TABLE II
FEATURE GROUPS AND COUNTS

| Feature Group | Features | Count |
|---|---|---|
| GWL lags ($\ell_1$–$\ell_4$) | Depth at prior observations | 4 |
| GWL change ($\Delta_1, \Delta_2$) | First differences | 2 |
| GWL rolling mean (3, 4 obs.) | Rolling averages | 2 |
| GWL rolling std (3 obs.) | Variability | 1 |
| Station history (mean, std) | Expanding statistics | 2 |
| Rainfall (7, 30, 90, 180, 365 d) | Accumulated totals | 5 |
| Temporal (year, month, quarter) | Calendar | 3 |
| Cyclical month (sin, cos) | Seasonal encoding | 2 |
| Station identifiers (name, block, ...) | Location | 6 |
| **Total** | | **27** |

The prediction horizon equals the number of days between consecutive observations, which ranges from 40 to 200 days (median 91 days across the filtered dataset; filter criterion: $[30, 200]$ days). No regression targets are created for the final observation at each station.

**Classification target.** The signed change between consecutive observations defines the depletion severity:

$$\Delta_t = \mathrm{GWL}_{t+1} - \mathrm{GWL}_t$$

where positive $\Delta_t$ denotes deepening (depletion) because GWL is measured as depth below ground level. The four classes are:

$$\text{Label} = \begin{cases} \text{No Depletion} & \Delta_t \le 0 \\ \text{Low} & 0 < \Delta_t \le 1\,\text{m} \\ \text{Moderate} & 1 < \Delta_t \le 3\,\text{m} \\ \text{High} & \Delta_t > 3\,\text{m} \end{cases}$$

These thresholds are empirical and fixed in metres, independent of observation interval length (40–200 days). Because a 1-m depth change over 40 days represents a different rate of depletion than over 200 days, interval-normalised thresholds (m/day) would be more physically meaningful. We retain the metre-based labels as originally defined but explicitly flag this as a limitation requiring hydrogeological validation before operational use. Class distribution in the full dataset: No Depletion 2,085 (49.8%), Low 847 (20.2%), Moderate 805 (19.2%), High 448 (10.7%).

### *B. System Architecture*

The pipeline (Fig. 1) is implemented as a decoupled FastAPI + React web application:

1) **Data ingestion**: Verified local CSV snapshots of NWDP datasets—CGWB quarterly groundwater (`3dacbc24`, `7b76b032`) and Karnataka daily rainfall—are ingested and filtered by `prepare_data.py` into district-level interim files.
2) **Feature engineering**: `build_features.py` joins rainfall rolling windows to GWL observations; `create_model_features.py` adds lag, change, and station-history features.
3) **Target construction**: `create_target.py` constructs regression and classification targets with horizon filtering.
4) **Model training**: `train_models.py` (regression) and `train_risk_model.py` (classification) apply chronological splits, fit pipelines (median imputation for numerics, mode imputation + OHE for categoricals), and save the best model as a `.joblib` artefact.
5) **Prediction**: `generate_all_predictions.py` loads the saved models and produces station-level prediction CSVs consumed by the dashboard.

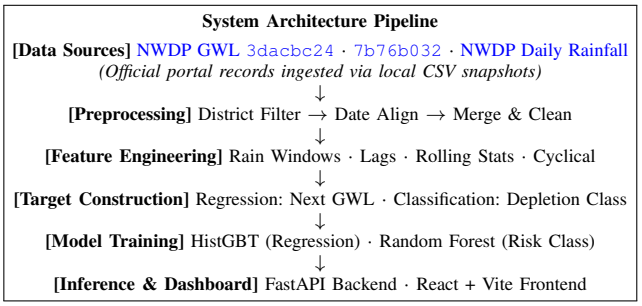



Fig. 1. Decoupled system architecture pipeline from official NWDP portal sources and local CSV training snapshots to web UI.

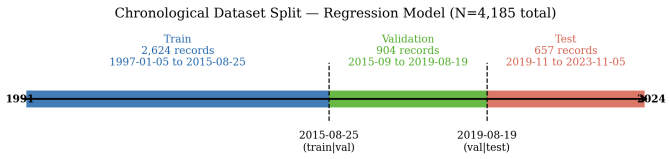



Fig. 2. Chronological dataset split for the regression model. Boundaries are determined by 70 %/85 % quantiles of unique observation dates; no station-date overlaps exist across partitions.

6) **Dashboard**: A React + Vite frontend served by a FastAPI backend presents predictions, confidence labels, and historical trends for exploratory use.

### C. Model Selection

Three ensemble regressors are benchmarked: **Random Forest** (500 trees, `max_features`=0.8), **Extra Trees** (600 trees, `max_features`=0.9), and **HistGradientBoostingRegressor** (HGB; `learning_rate`=0.05, `max_iter`=500, `l2_regularization`=0.1). All models are wrapped in scikit-learn Pipelines with a `ColumnTransformer` (median imputer for numerics, mode imputer + one-hot encoder for categoricals). HGB is selected as the final regression model on the basis of lowest validation-set RMSE. For classification, the same three estimators are compared as classifiers; Random Forest achieves highest validation accuracy and is selected.

Hyperparameters were set by manual exploration guided by validation performance; no exhaustive grid search was conducted. All random states are fixed at 42 for reproducibility.

## V. Experimental Setup

### A. Chronological Data Split

The dataset is sorted by date and split chronologically by unique monitoring dates, preventing data leakage across splits. The split proportions are 70 %/15 %/15 % of unique dates (Fig. 2):

- **Train**: 2,624 records, 1997-01-05 to 2015-08-25
- **Validation**: 904 records, 2015-11-05 to 2019-08-19
- **Test**: 657 records, 2019-11-11 to 2023-11-05

The final regression model is retrained on train + validation combined before evaluation on the held-out test set.

*Split audit*: Regression uses 2,624/904/657 records (70 %/15 %/15 % unique dates); classification purges 113 boundary transitions (`next_date` ≤ `split_end`), yielding 2,569/846/657 records. Both test on identical 657 records.

### B. Uncertainty Estimation

Point-prediction uncertainty is evaluated via observation-level empirical **bootstrap resampling** (i.i.d., 1,000 iterations, seed 42) on the 657 test records, yielding empirical 95% confidence intervals of [1.77, 2.10] m for MAE, [2.61, 3.32] m for RMSE, and [0.423, 0.639] for $R^2$. Resampling captures test variance without assuming block temporal structure or epistemic uncertainty.

### C. Feature Importance Method

Feature importance is computed by **permutation importance** on the validation set (5 repetitions, random seed 42). Importance is defined as the decrease in $R^2$ when each feature column is randomly shuffled. Negative values indicate features whose permutation slightly improves performance (noise features); these are reported transparently rather than suppressed.

### D. Baselines

**Regression baselines** evaluated on the same 657 test records:

- *Mean-target*: predicts the training-set mean GWL for every observation.
- *Persistence*: predicts the current observed GWL as the next GWL (*no-change* model).

**Classification baselines**:

- *Majority-class*: always predicts "No Depletion".
- *Stratified dummy*: samples from the training-set class distribution (seed 42).

## VI. Results

All metrics reported here are computed from saved prediction artefacts (`models/test_predictions.csv` and `models/risk_test_predictions.csv`) on the chronologically held-out test set (657 records, November 2019 – November 2023).

### A. Regression Results

Table III shows test-set performance. The HistGBT model reduces MAE by 43.4% relative to the mean-target baseline and by 9.8% relative to the persistence baseline. The $R^2$ of 0.547 indicates that roughly 55% of GWL variance in the test period is explained by the model. The relatively modest $R^2$ reflects the wide 40–200-day prediction horizon and the inherent variability of manually monitored sparse wells.

TABLE III
Regression Test-Set Performance (N = 657, Nov 2019–Nov 2023)

| Model | MAE (m) | RMSE (m) | $R^2$ |
|---|---|---|---|
| Mean-target baseline | 3.43 | 4.39 | — |
| Persistence baseline | 2.15 | 3.39 | — |
| **HistGBT (ours)** | **1.94** | **2.96** | **0.547** |

All metrics computed from saved artefacts.
Units: metres below ground level (mbgl).

Fig. 3 shows the actual-versus-predicted scatter plot and the residual distribution. The model tends to underpredict at

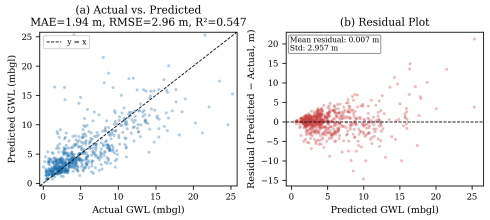



Fig. 3. Regression results on the held-out test set (N = 657). (a) Actual vs. predicted GWL (mbgl) with y=x reference. (b) Residual plot; mean residual −0.01 m indicates negligible systematic bias.

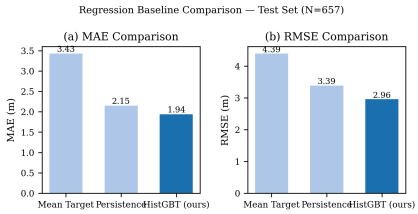



Fig. 4. Regression baseline comparison on the test set. HistGBT reduces MAE by 43% vs. mean-target baseline and by 10% vs. the persistence baseline.

extreme high depths (> 15 mbgl), reflecting the scarcity of such observations in the training period.

### B. Classification Results

Table IV reports classification performance for the Random Forest model. The model outperforms both baselines on accuracy and macro-F1. High-risk classification (F1 = 0.33) is the weakest, reflecting the class imbalance (only 80 test examples, 12.2%) and the difficulty of detecting extreme depletion events in quarterly data.

TABLE IV
CLASSIFICATION TEST-SET PERFORMANCE (N = 657, NOV 2019–NOV 2023)

| Model | Accuracy | Macro F1 | Wt. F1 |
|---|---|---|---|
| Majority-class baseline | 0.458 | 0.157 | — |
| Stratified dummy (seed=42) | 0.304 | 0.253 | — |
| **Random Forest (ours)** | **0.528** | **0.457** | **0.529** |

TABLE V
PER-CLASS PERFORMANCE — RANDOM FOREST (TEST SET)

| Class | Prec. | Recall | F1 | Support |
|---|---|---|---|---|
| No Depletion | 0.70 | 0.69 | 0.70 | 301 |
| Low | 0.38 | 0.39 | 0.38 | 129 |
| Moderate | 0.42 | 0.42 | 0.42 | 147 |
| High | 0.33 | 0.33 | 0.33 | 80 |
| Macro avg | 0.46 | 0.46 | 0.46 | 657 |
| Weighted avg | 0.53 | 0.53 | 0.53 | 657 |

Fig. 5 presents the confusion matrix (counts and row-normalised fractions). The dominant error mode for No Depletion (301 actual) is misclassification as Moderate (34 cases) or High (30 cases), which would constitute false alarms in a screening application. For High depletion (80 actual), 48 cases are misclassified—32 as Moderate and 16 as No Depletion—representing missed detections that would be operationally consequential.

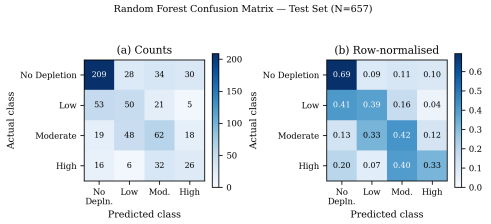



Fig. 5. Random Forest confusion matrix — test set (N = 657). (a) Raw counts; (b) row-normalised fractions. Diagonal entries represent correct classifications.

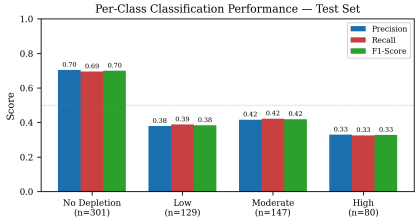



Fig. 6. Per-class precision, recall, and F1-score for the Random Forest classifier. Support counts shown below class labels.

### C. Feature Importance

Permutation importance (Fig. 7) identifies current GWL (`groundwater_level_m`, importance 0.592) and the 4-observation rolling mean (`gw_rolling_mean_4`, 0.328) as the most influential features. Cyclical month encoding and station historical mean follow. Short-window rainfall features (7-day: 0.011) carry modest positive importance, while longer-window rainfall features (90, 180, 365 days) show near-zero or slightly negative importance on the validation set. This does not imply that long-term rainfall is physically unimportant; it may reflect collinearity with GWL history, irregular rainfall data coverage, or the limited discriminative power of district-averaged rainfall at the station scale. Caution is warranted in interpreting negative importance values as evidence of harmful features.

## VII. DISCUSSION, LIMITATIONS, AND THREATS TO VALIDITY

### A. Key Findings

The HistGBT regressor meaningfully reduces prediction error relative to both the mean-target and persistence baselines, suggesting that GWL history and station characteristics carry predictive signal beyond a no-change assumption. The four-class depletion classifier achieves modest but above-chance performance, outperforming both baselines on macro-F1 by factors of 2.9× and 1.8× respectively.

### B. Limitations

1) **Sparse station network.** 92 manual observation wells across Mysuru district (approximately 6,645 km$^2$) provide an average density of roughly 1 station per 72 km$^2$. Unmonitored areas, particularly those with deep confined aquifers or significant extraction, are not represented.
2) **Variable prediction horizon.** The model predicts one observation ahead regardless of the inter-observation

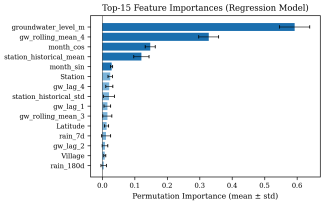



Fig. 7. Top-15 permutation feature importances (mean $\pm$ std) for the regression model. Importance is defined as the decrease in $R^2$ upon random permutation of each feature.

gap (40–200 days). Fixed-metre risk thresholds conflate different rates of change across intervals; hydrogeologically, a 3-m change over 40 days is far more alarming than the same change over 200 days.

3) **Missing rainfall.** Gaps in the NWDP daily rainfall series (including December 2023) mean that rolling accumulation features carry partial-window uncertainty near gap boundaries. Missing rainfall is retained as `NaN` (not imputed to zero), but propagation through rolling windows with `min_periods=1` means that some features silently cover fewer days than their window name implies.
4) **Empirical risk thresholds.** The Low/Moderate/High boundaries (1 and 3 m) are heuristic and not validated against CGWB's official semi-critical/critical/over-exploited classifications, which are based on multi-year extraction-recharge balances rather than observation-to-observation changes.
5) **High-risk detection.** F1 of 0.33 for the High class means that roughly two in three High-risk events are not correctly identified. The model is not reliable for early warning of extreme depletion without further improvement.
6) **Rainfall feature influence.** Most rainfall permutation-importance values are near zero or negative, particularly for long windows. This may reflect collinearity, district-averaged aggregation losing local signal, or the lagged relationship between rainfall and shallow versus deep aquifer response.
7) **No temporal cross-validation.** A single temporal hold-out is used. Walk-forward cross-validation would provide more robust performance estimates but was not feasible given dataset size.
8) **Dashboard is exploratory.** The web dashboard presents model outputs for visual exploration only. It does not provide calibrated probabilities, uncertainty bands, or validation against real-time measurements. It is not suitable for operational groundwater management decisions.

### C. Threats to Validity

**Internal validity**: The strictly chronological split with no overlap between train, validation, and test periods prevents temporal leakage. All lag and rolling features use only information available before the prediction date. **External validity**: Results may not generalise to other districts, aquifer types (fractured rock vs. alluvial), or monitoring protocols different from CGWB quarterly campaigns. **Construct validity**: The "next observation" horizon is driven by monitoring schedule rather than any operationally meaningful lead time. Decisions based on predictions require knowledge of when the next monitoring visit is planned.

## VIII. Conclusion and Future Work

We presented an exploratory machine-learning system for GWL prediction and depletion-risk classification at CGWB monitoring wells in Mysuru district, Karnataka. Using 27+ years of quarterly observations from 92 stations and district-averaged NWDP daily rainfall (ingested from local snapshots), a Historical Gradient Boosting Regressor achieves MAE = 1.94 m and $R^2 = 0.547$ on 657 chronologically held-out test records (2019–2023), reducing prediction error by 43% versus the mean baseline. A companion Random Forest classifier achieves 52.8% accuracy and macro-F1 of 0.457 for four-class depletion severity, outperforming both majority-class and stratified baselines.

Key limitations include variable prediction horizons with fixed-metre risk thresholds, sparse monitoring coverage, partial rainfall data, and low recall on the rare High-risk class. The system is presented as an exploratory decision-support prototype rather than an operational early-warning tool. Future work should: (1) validate risk thresholds against official CGWB groundwater-resource assessments; (2) incorporate aquifer-type and lithology covariates; (3) explore calibrated probabilistic outputs (e.g., conformal prediction); (4) evaluate interval-normalised change rates as alternative classification targets; and (5) extend the framework to adjacent districts with similar monitoring infrastructure.

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